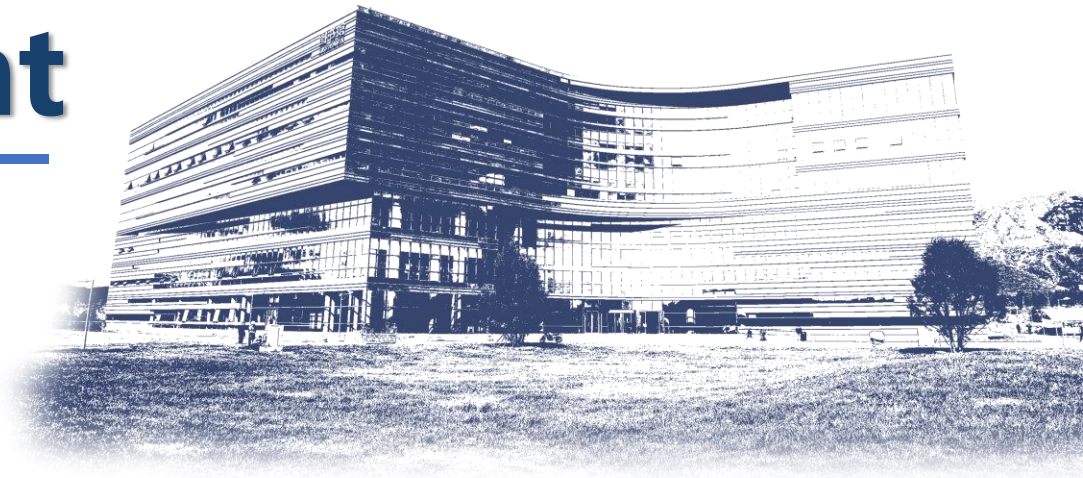
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3.Project Management

System Analysis and Design



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Learning Objectives

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- Understand how projects are initiated and selected, define a business problem, and determine the feasibility of a proposed project
- Evaluate hardware and software and the way it supports human interactions with technology
- Forecast and analyze tangible and intangible costs and benefits
- Manage a project by preparing a budget, creating a work breakdown structure, scheduling activities, and controlling the schedule and costs
- Build and manage a project team
- Professionally write and present an effective systems proposal, concentrating on both content and design

Project Management Fundamentals

- Project initiation
- Determining project feasibility
- Activity planning and control
- Project scheduling
- Managing systems analysis team members

Major Topics

- Project initiation
- Determining feasibility
- Determining resources
- Activity planning and control
  - Gantt charts
  - PERT diagrams
- Managing analysis and design activities
- The agile approach

3.1 Project Initiation(项目启动)

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- Problems in the organization
  - Problems that lend themselves to systems solutions
- Opportunities for improvement
  - Caused through upgrading, altering, or installing new systems

3.1.1 Checking Output

Observing employee behavior, and listening to feedback are all ways to help the analyst pinpoint systems problems and opportunities.

To Identify Problems	Look for These Specific Signs:
Check output against performance criteria.	Too many errors Work completed slowly Work done incorrectly Work done incompletely Work not done at all
Observe behavior of employees.	High absenteeism High job dissatisfaction High job turnover
Listen to external feedback from: Vendors and service providers. Customers. Suppliers.	Complaints Suggestions for improvement Loss of sales Lower sales

3.1.2 Problem Definition

- Problem statement
  - Paragraph or two stating the problem or opportunity
- Issues
  - Independent pieces pertaining to the problem or opportunity
- Objectives
  - Goals that match the issues point-by-point
- Requirements
  - The things that must be accomplished along with the possible solutions, and constraints, that limit the development of the system
- Use the problem definition to create a preliminary test plan

Problem Definition: Identifying Issues

- Find a number of points that may be included in one issue
- State the objective
- Determine the relative importance of the issues or objectives
- Identify which objectives are most critical

3.1.3 Selection of Projects

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- Backing from management
- Appropriate timing of project commitment
- Possibility of improving attainment of organizational goals
- Practical in terms of resources for the system analyst and organization
- Worthwhile project compared with other ways the organization could invest resources

Selection of Projects: Improving Attainment of Organizational Goals

- Improving corporate profits
- Supporting the competitive strategy of the organization
- Improving cooperation with vendors and partners
- Improving internal operations support
- Improving internal decision support so that decisions are more effective
- Improving customer service
- Increasing employee morale

Defining Objectives

Many possible objectives exist including:

- **Speeding up a process**
- **Streamlining a process**
- **Combining processes**
- **Reducing errors in input**
- **Reducing redundant storage**
- **Reducing redundant output**
- **Improving system and subsystem integration**

3.2 Determining Feasibility

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- Assessed in three principal ways
  - Operationally
  - Technically
  - Economically

3.2.1 The Three Key Elements of Feasibility Include Technical, Economic, and Operational Feasibility

The Three Key Elements of Feasibility

- Technical Feasibility
  - Add on to present system
  - Technology available to meet users' needs
- Economic Feasibility
  - Systems analysts' time
  - Cost of systems study
  - Cost of employees' time for study
  - Estimated cost of hardware
  - Cost of packaged software or software development
- Operational Feasibility
  - Whether the system will operate when put in service
  - Whether the system will be used

Technical Feasibility

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- Can current technical resources be upgraded or added to in a manner that fulfills the request under consideration?
- If not, is there technology in existence that meets the specifications?

Economic Feasibility

- Economic feasibility determines whether value of the investment exceeds the time and cost
- Includes:
  - Analyst and analyst team time
  - Business employee time
  - Hardware
  - Software
  - Software development

Operational Feasibility

- Operational feasibility determines if the human resources are available to operate the system once it has been installed
- Users that do not want a new system may prevent it from becoming operationally feasible

3.2.2 Estimating Workloads

- **Systems analysts formulate numbers that represent both current and projected workloads for the system so that any hardware obtained will possess the capability to handle current and future workloads**

Comparisons of Workloads between Existing and Proposed Systems

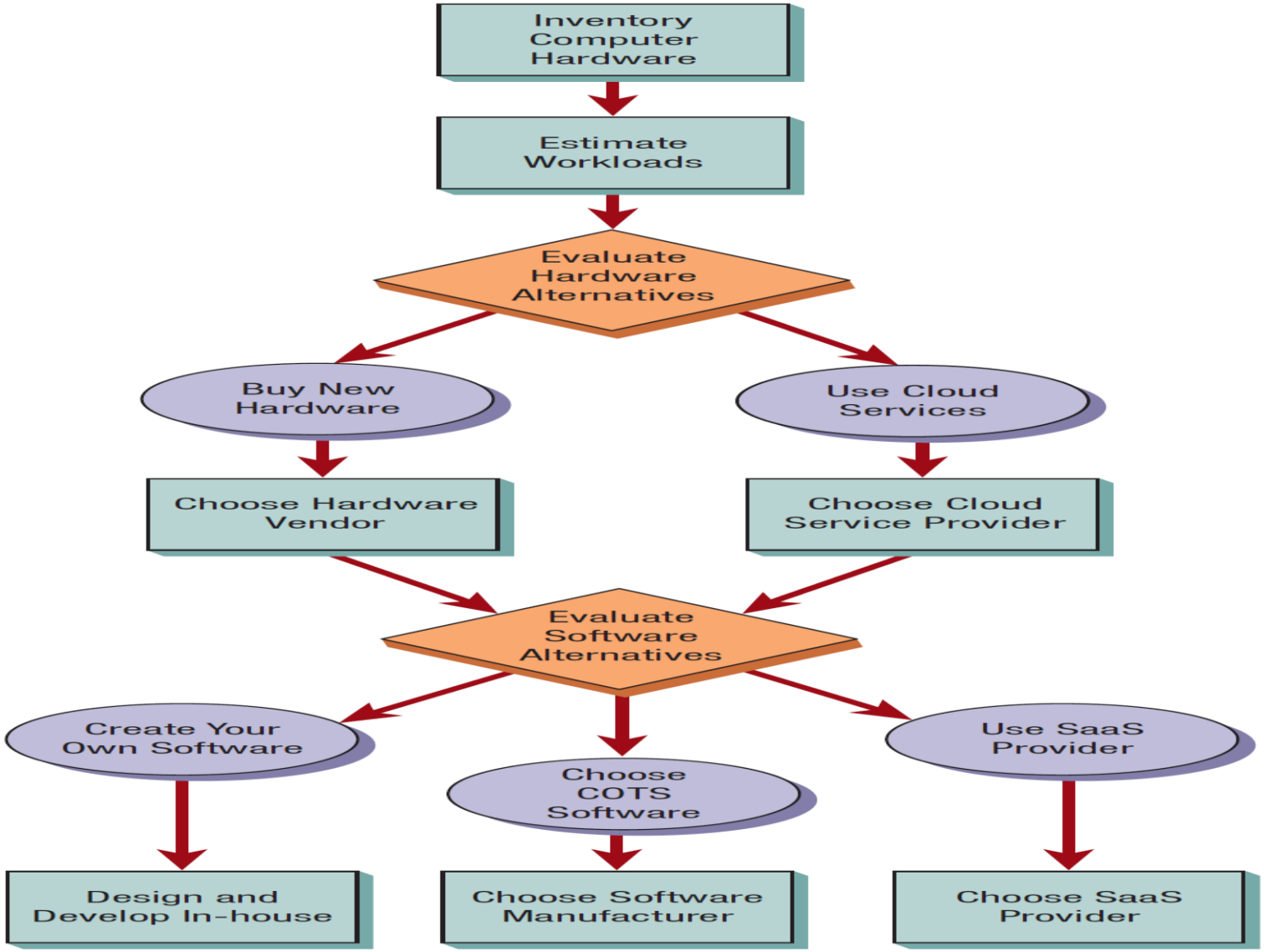
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	Existing System	Proposed System
Task	Compare performance of distribution warehouses by running the summary program.	Compare performance of distribution warehouses on the Web-based dashboard.
Method	Computer programs are run when needed; processing is done from the workstation.	Updates occur immediately; processing is done online.
Personnel	Distribution manager	Distribution manager
When and how	Daily: Enter shipments on Excel spreadsheet; verify accuracy of spreadsheet manually; and then write files to backup media. Monthly: Run program that summarizes daily records and prints report; get report and make evaluations.	Daily: Enter shipments on the Web-based system using automatically backed up drop-down boxes. Data are to remote location. Monthly: Compare warehouses online using the performance dashboard; print only if needed.
Human time requirements	Daily: 20 minutes Monthly: 30 minutes	Daily: 10 minutes Monthly: 10 minutes
Computer time requirements	Daily: 20 minutes Monthly: 30 minutes	Daily: 10 minutes Monthly: 10 minutes

3.3 Ascertaining Hardware and Software Needs

- Steps used to determine hardware and software needs:
  - Inventory computer hardware currently available
  - Estimate current and future system workloads
  - Evaluate available hardware and software
  - Choose the vendor
  - Acquire the computer equipment

Steps in Choosing Hardware and Software



3.3.1 Inventorying Computer Hardware

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- Type of equipment
- Operation status of the equipment
- Estimated age of equipment
- Projected life of equipment
- Physical location of equipment
- Department or person responsible for equipment
- Financial arrangement for equipment

3.3.2 Evaluating Hardware

- Time required for average transactions
- Total volume capacity of the system
- Idle time of the CPU or network
- Size of memory provided

People That Evaluate Hardware

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- Management
- Users
- Systems analysts

Acquisition of Computer Equipment

- **Purchasing**
- **Using Cloud Services**

3.3.3 Available Cloud Services

- Available cloud services may include:
  - Web hosting
  - Email hosting
  - Application hosting
  - Backup
  - Storage and processing of databases
  - Archiving
  - Ecommerce

Three Main Categories of Cloud Computing

- Software as a Service (SaaS)
- Infrastructure as a Service (IaaS)
- Platform as a Service (PaaS)

Strategic Cloud Computing Decisions

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- Decisions on cloud computing can first be addressed on a strategic level
- Business should focus on:
  - Define a high-level business case that focuses on high-level benefits
  - Define core requirements
  - Define core technologies for the enterprise

Benefits of Cloud Computing

- Less time spent maintaining legacy systems
- May be simpler to acquire IT services
- Scalable applications that can grow by adding more cloud resources
- Consistency across multiple platforms
- Capital is not tied up

Drawbacks of Cloud Computing

- Loss of control of data stored in the cloud
- Potential security threats to data that is not stored on premises
- Reliability of the Internet as a platform

Purchasing Hardware or Using Cloud Services Advantages and Disadvantages

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	Advantages	Disadvantages
Buying computer Hardware	Full control over hardware and software Often cheaper in long run Provides tax advantages through Depreciation	Initial cost is high Risk of obsolescence Risk of being stuck if choice was wrong Full responsibility for operation and maintenance
Using cloud Services	Maintenance and upgrades performed by provider Ability to change software and hardware rapidly Scalable—can grow quickly Consistent over multiple platforms No capital is tied up	Company doesn't control its own data Data security is at risk Reliability risks of the Internet platform Proprietary APIs and software may make switching providers difficult

3.3.4 Evaluating Vendor（厂商） Support

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- Hardware support
- Software support
- Installation and training support
- Maintenance support

Guidelines for Vendor Selection

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Vendor Services	Specifics Vendors Typically Offer
Hardware Support	Full line of hardware Quality products Warranty
Software Support	Complete software needs Custom programming Warranty
Installation and Training	Commitment to schedule In-house training Technical assistance
Maintenance	Routine maintenance procedures Specified response time in emergencies Equipment loan while repair is being done
Cloud Services	Web hosting Email hosting Data storage
Software as a Service	Automatic software upgrades Support services Security and antivirus protection

3.3.5 BYOD and BYOT

- BYOD: Bring your own device（自带设备）
- BYOT: Bring your own technology（自带技术）
- Employee uses their own device access corporate networks, data, and services remotely

Benefits of BYOD and BYOT

- Building employee morale
- Potential for lowering the initial cost hardware purchase
- Facilitating remote, around-the-clock access to corporate computer networks
- Building on a familiar user interface to access corporate computing services, applications, databases, and storage

Drawbacks of BYOD and BYOT

- Security risks posed by untrained users
- Loss of the device
- Theft of the device and its data
- Unauthorized access to corporate networks using personal mobile devices

3.3.6 Software Alternatives

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- Created custom software（定制软件）
- Purchased as COTS (commercial off-the-shelf) software（商用成品）
- Provided by an software as a service (SaaS)

Software Alternatives

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	Advantages	Disadvantages
Creating Custom Software	Specific response to specialized business needs Innovation may give firm a competitive advantage In-house staff available to maintain software Pride of ownership	May be significantly higher initial cost compared to C OTS software or ASP Necessity of hiring or working with a development team Ongoing maintenance
Purchasing C OTS Packages	Refined in the commercial world Increased reliability Increased functionality Often lower initial cost Already in use by other firms Help and training comes with software	Programming focused; not business focused Must live with the existing features Limited customization Uncertain financial future of vendor Less ownership and commitment
Using SaaS	Organizations that do not specialize in information systems can focus on what they do best (their strategic mission) There is no need to hire, train, or retain a large IT staff There is no expenditure of employee time on nonessential IT tasks	Loss of control of data, systems, IT employees, and schedules Concern over the financial viability and long-run stability of the SaaS provider Security, confidentiality, and privacy concerns Loss of potential strategic corporate advantage regarding innovativeness of applications

Software Evaluation

- Performance effectiveness
- Performance efficiency
- Ease of use
- Flexibility
- Quality of documentation
- Manufacturer support

Guidelines for Evaluating Software

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Software Requirements	Specific Software Features
Performance Effectiveness	Able to perform all required tasks Able to perform all tasks desired Well-designed display screens Adequate capacity
Performance Efficiency	Fast response time Efficient input Efficient output Efficient storage of data Efficient backup
Ease of Use	Satisfactory user interface Help menus available “Read Me” files for last-minute changes Flexible interface Adequate feedback Good error recovery
Flexibility	Options for input Options for output Usable with other software
Quality of Documentation	Good organization Adequate online tutorial Website with FAQ
Manufacturer Support	Technical support hotline Newsletter/email Website with downloadable product updates

Identifying and Forecasting Costs and Benefits

- Judgment methods
  - Estimates from the sales force
  - Surveys to estimate customer demand
  - Delphi studies
  - Creating scenarios
  - Drawing historical analogies

- If historical data are available
  - Conditional:
    - There is an association among variables in the model
  - Unconditional:
    - Do not need to find or identify any relationships

Estimation of Trends

- **Graphical judgment**
- **Moving averages**

3.4 Identifying Benefits and Costs (效益和成本)

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- Tangible benefits are advantages measurable in dollars through the use of the information system
- Intangible benefits are difficult to measure
- Tangible costs are accurately projected by the systems analyst and accounting personnel
- Intangible costs are difficult to estimate and may not be known

Tangible Benefits（有形效益）

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- Advantages measurable in dollars that accrue to the organization through the use of the information system
- Examples:
  - Increase in the speed of processing
  - Access to otherwise inaccessible information
  - Access to information on a more timely basis
  - The advantage of the computer's superior calculating power
  - Decreases in the amount of employee time needed to complete specific tasks

Intangible Benefits（无形效益）

- Intangible benefits are benefits from use of the information system that are difficult to measure
- Examples:
  - Improving the decision-making process
  - Enhancing accuracy
  - Becoming more competitive in customer service
  - Maintaining a good business image
  - Increasing job satisfaction

Tangible Costs（有形成本）

- Those that can be accurately projected by systems analysts and the business' accounting personnel
- Examples:
  - Cost of equipment
  - Cost of resources
  - Cost of systems analysts' time
  - Cost of programmers' time
  - Employees' salaries

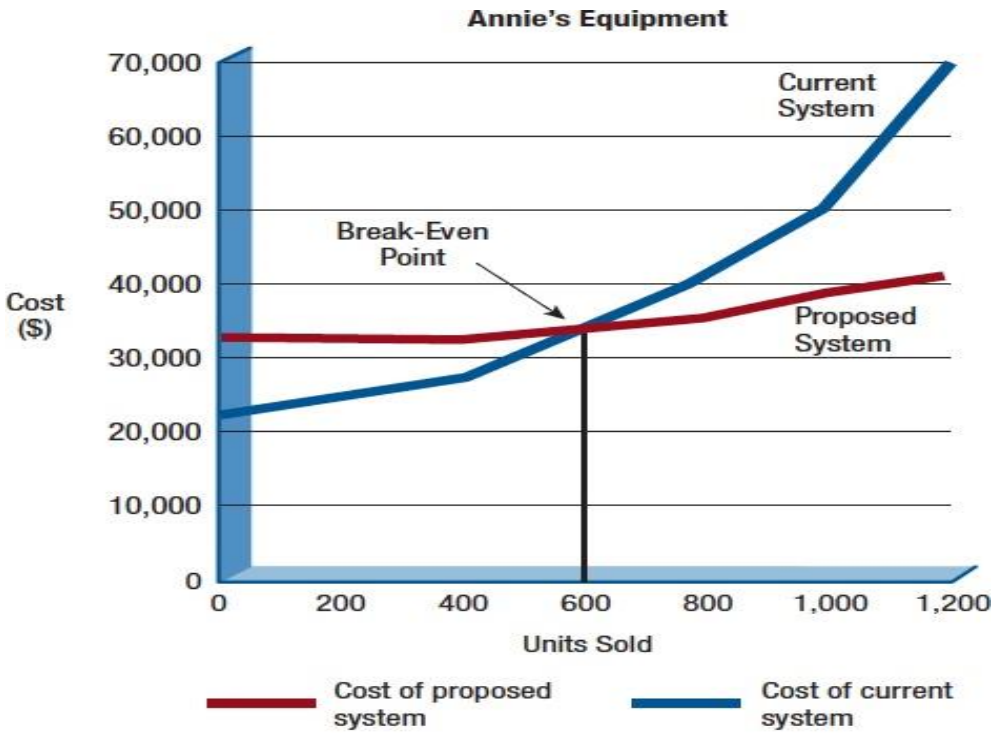
Intangible Costs

- Those that are difficult to estimate and may not be known
- Examples:
  - Losing a competitive edge
  - Losing the reputation of being first
  - Declining company image
  - Ineffective decision making

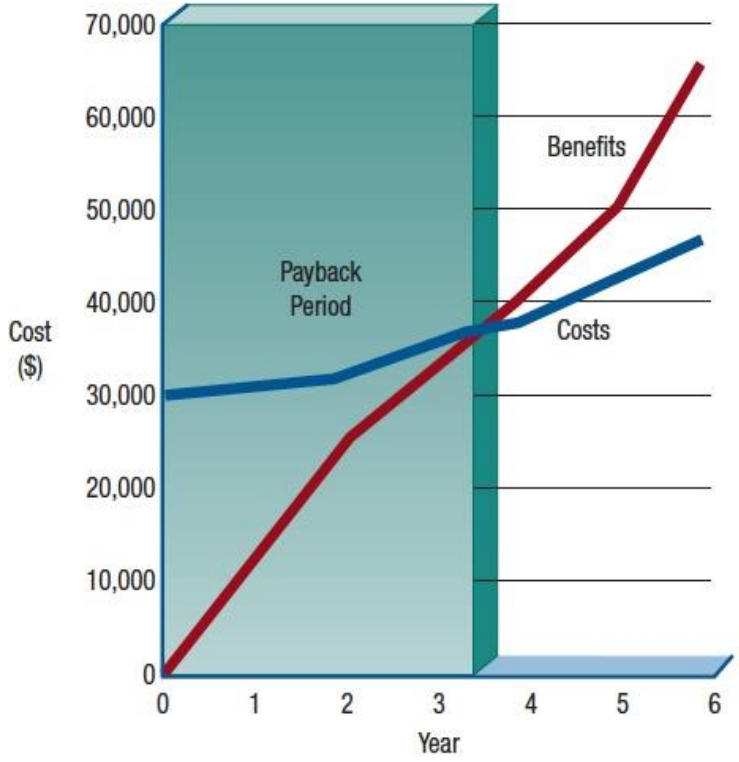
Comparing Costs and Benefits

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- **Break-even analysis**
- **Payback**



Break-Even Analysis



Cumulative benefits from proposed system

Cumulative costs of proposed system

Year	Cost ($)	Cumulative Costs ($)	Benefits ($)	Cumulative Benefits ($)
0	30,000	30,000	0	0
1	1,000	31,000	12,000	12,000
2	2,000	33,000	12,000	24,000
3	2,000	35,000	8,000	32,000
4	3,000	38,000	8,000	40,000
5	4,000	42,000	10,000	50,000
6	4,000	46,000	15,000	65,000

Payback Analysis Showing a Payback Period of Three and a Half Years

3.5 Work Breakdown Structure

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- Often a project needs to be broken down into smaller tasks or activities
- These tasks together make up a work breakdown structure (WBS)

Work Breakdown Structure Properties

- Each task or activity contains one deliverable, or tangible outcome, from the activity
- Each task can be assigned to a single individual or a single group
- Each task has a responsible person monitoring and controlling performance

Developing a Work Breakdown Structure (WBS)

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- Decomposition, starting with large ideas, then breaking them down into manageable activities
- Product oriented, building a website can be broken down into many parts
- Process-oriented, emphasizes the importance of each phase

1.0	**Project Initiation**
1.1	Assemble and welcome project team
1.2	Conduct historical research about the business
1.3	Discuss objectives with client
2.0	**Early Planning Phase**
2.1	Investigate feasibility
2.2	Consider a make vs. buy decision
2.3	Develop a work breakdown structure
2.4	Provide time estimates
2.5	Develop a project schedule
2.6	Calculate cost estimates
2.7	Prepare project proposal to client
2.8	Present proposal to client
3.0	**Develop supporting plans**
3.1	Develop a quality management plan
3.2	Identify risks and build a risk management plan
3.3	Describe a communications plan
3.4	Develop a procurement plan
4.0	**Analysis**
4.1	Conduct interviews of key personnel
4.2	Administer questionnaires
4.3	Read company reports
4.4	Analyze data flow
5.0	**Design**
5.1	Build prototype website
5.2	Obtain reactions from client
5.3	Modify prototype website
5.4	Seek final recommendations from client
5.5	Complete website
6.0	**Launch**
6.1	Create training manual
6.2	Document website features and logic
6.3	Present final website to client

A Sample Work Breakdown Structure (WBS)

Time Estimation Techniques (时间估计技术)

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- Relying on experience
- Using analogies
- Using three-point estimation
- Identifying function points
- Using time estimation software

Function Point Analysis

- Takes the five main components of a computer system and rates them in terms of complexity:
  - External inputs
  - External outputs
  - External queries
  - Internal logical files
  - External interface files

Beginning to Plan a Project by Breaking It into Three Major Activities

Phase	Activity
Analysis	Data gathering Data flow and decision analysis Proposal preparation
Design	Data entry design Input design Output design Data organization
Implementation	Implementation Evaluation

Break apart the major activities into smaller ones.

Refining the Planning and Scheduling of Analysis Activities by Adding Detailed Tasks and Establishing the Time Required to Complete the Tasks

Activity	Detailed Activity	Weeks Required
Data gathering	Conduct interviews	3
	Administer questionnaires	4
	Read company reports	4
	Introduce prototype	5
	Observe reactions to prototype	3
Data flow and decision analysis	Analyze data flow	8
Proposal preparation	Perform cost-benefit analysis	3
	Prepare proposal	2
	Present proposal	2

Break these down further,

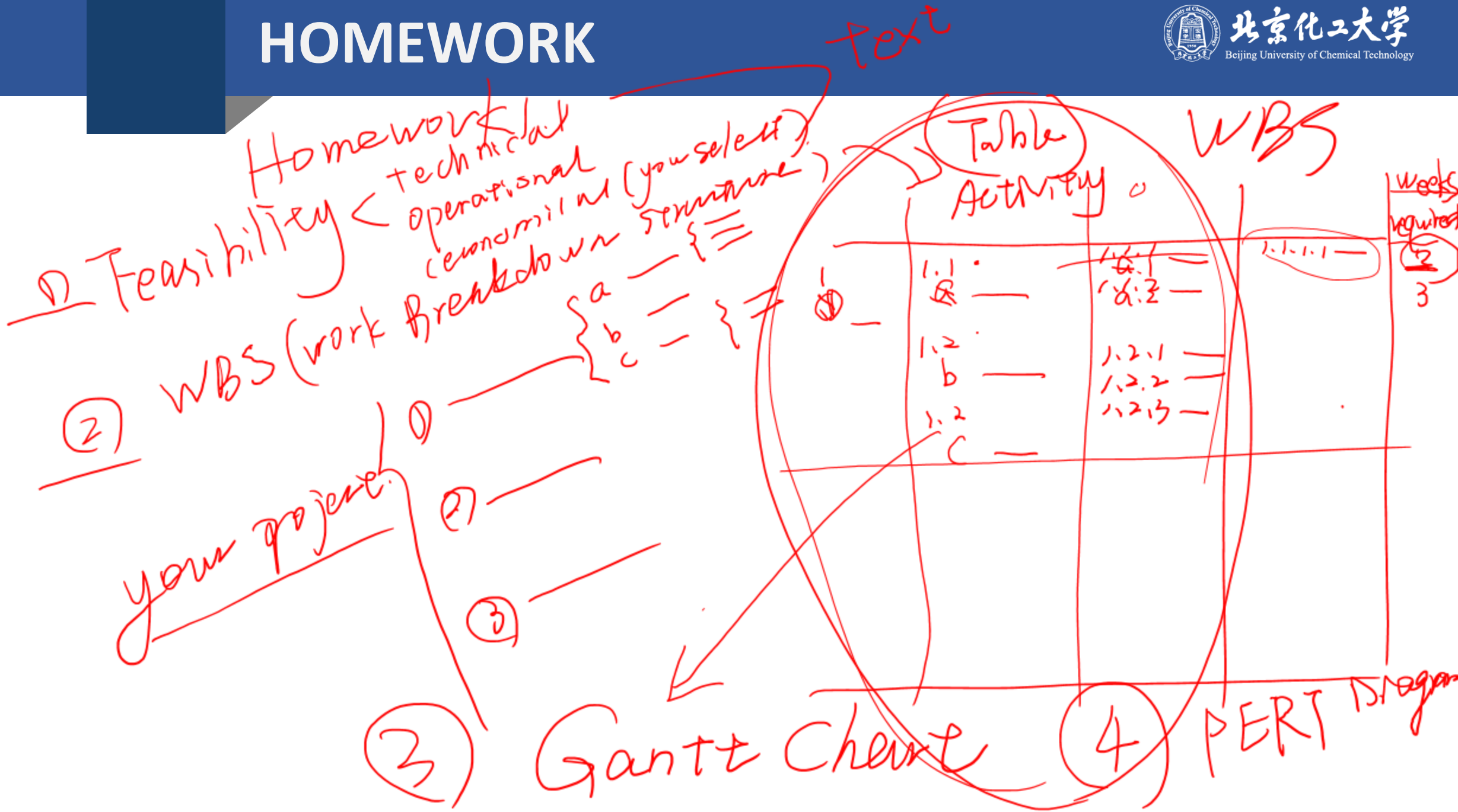
then estimate time required.

HOMEWORK

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HOMEWORK

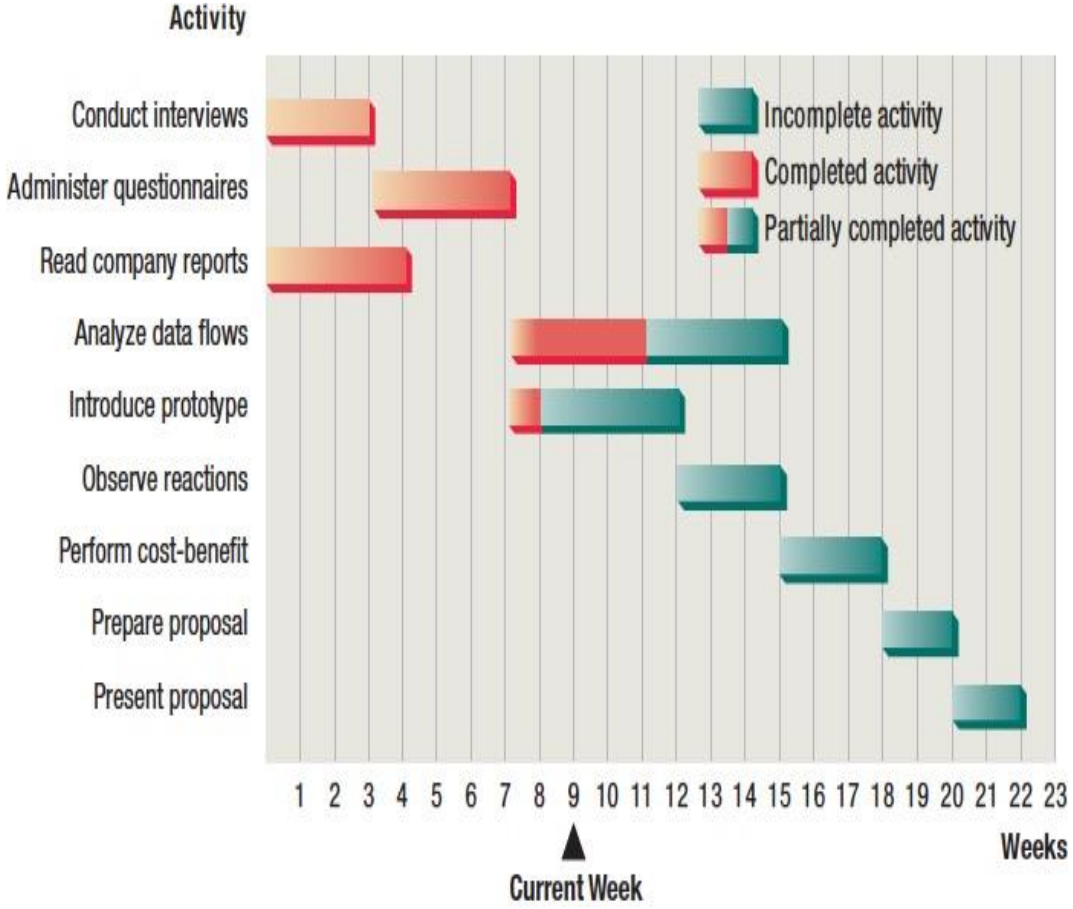
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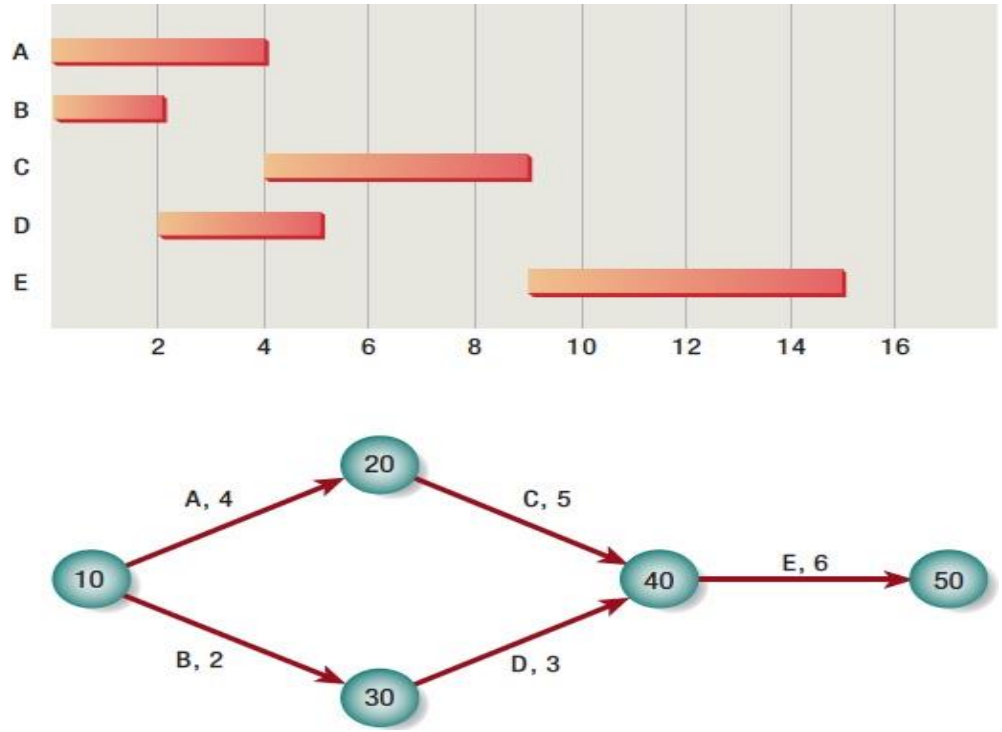
3.6 Project Scheduling

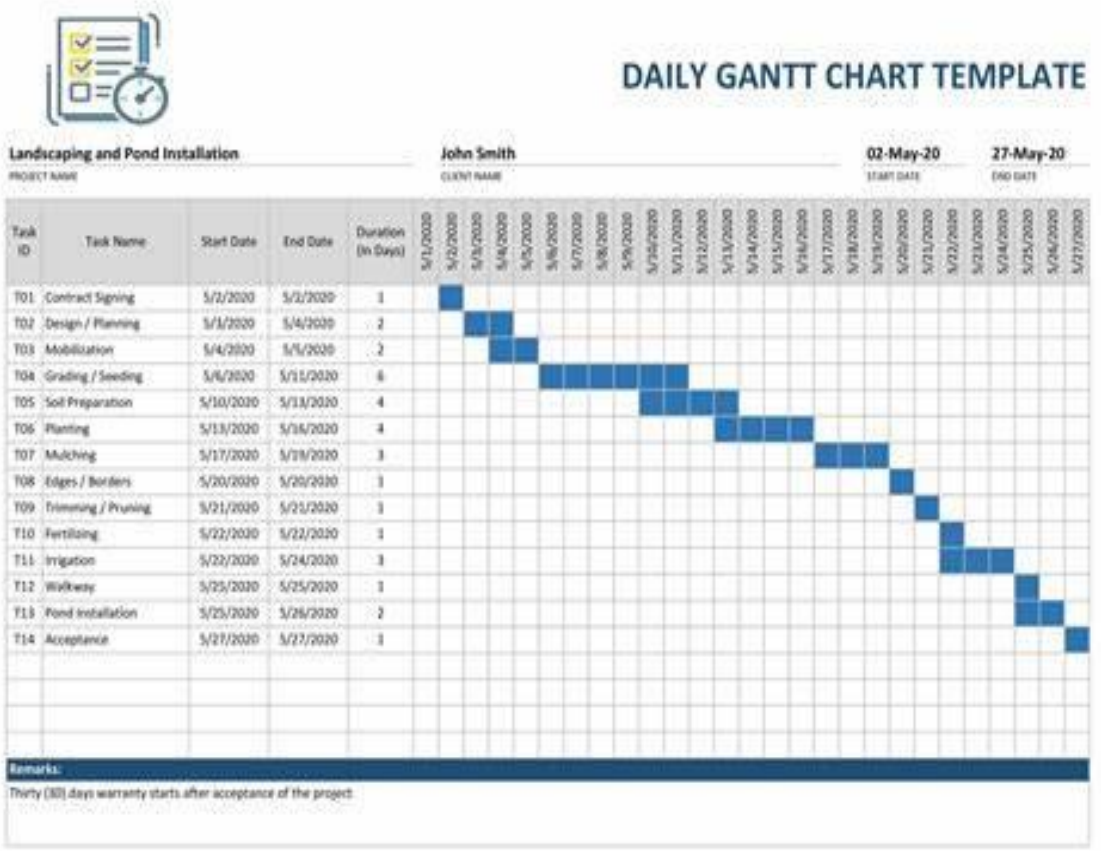
- Gantt Charts
  - Simple
  - Lends itself to end user communication
  - Drawn to scale
- PERT（程序评审Program evaluation and review） diagrams
  - Useful when activities can be done in parallel

Using a Two-Dimensional Gantt Chart for Planning Activities That Can Be Accomplished in Parallel



A Gantt Chart Compared with a PERT Diagram (Network Diagram) for Scheduling Activities



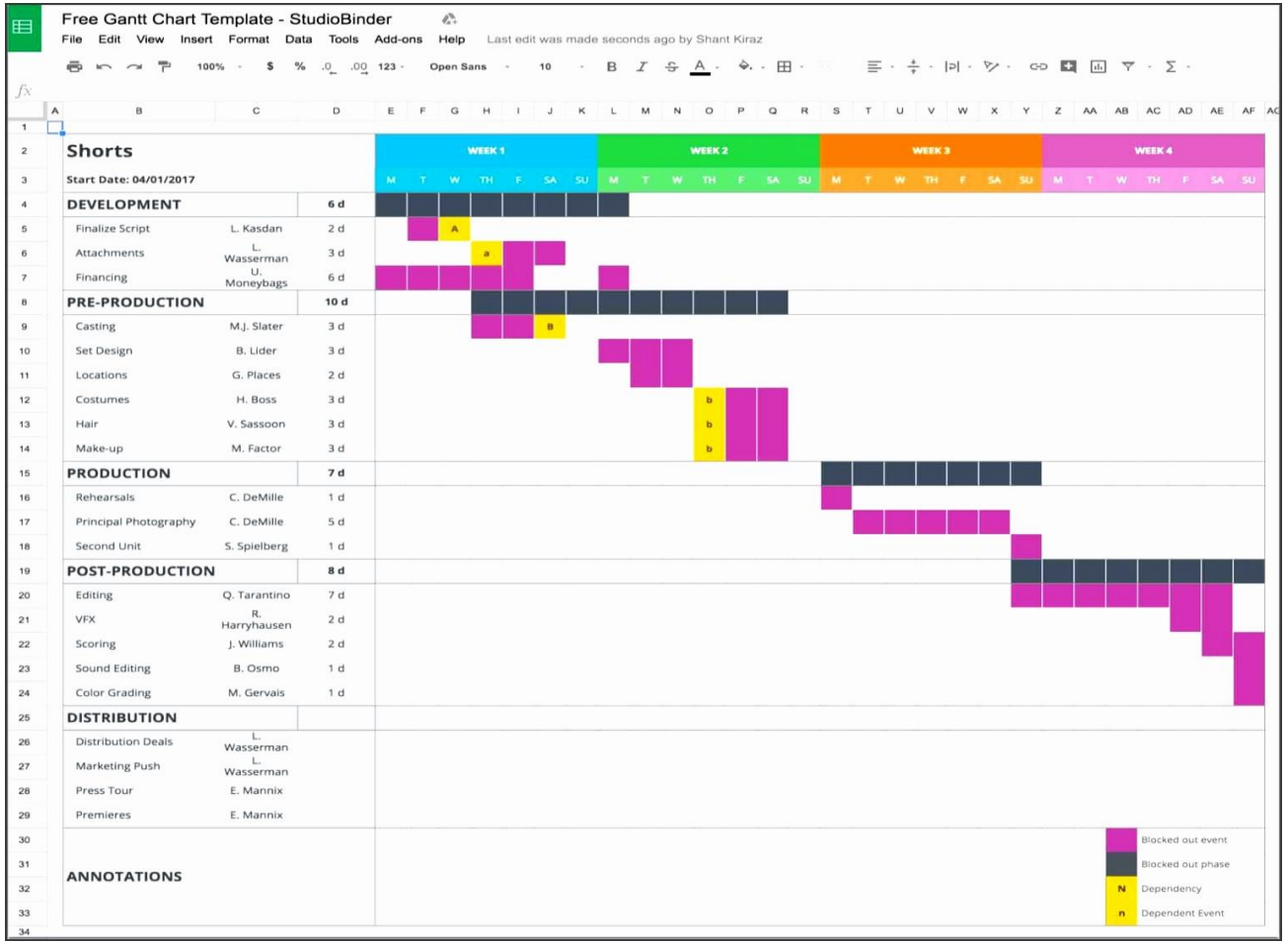


DAILY GANTT CHART TEMPLATE

Landscaping and Pond Installation (PROJECT NAME) | **John Smith** (CLIENT NAME) | **02-May-20** (START DATE) | **27-May-20** (END DATE)

Task ID	Task Name	Start Date	End Date	Duration (in Days)
T01	Contract Signing	5/2/2020	5/2/2020	1
T02	Design / Planning	5/3/2020	5/4/2020	2
T03	Mobilization	5/4/2020	5/5/2020	2
T04	Grading / Seeding	5/6/2020	5/11/2020	6
T05	Soil Preparation	5/10/2020	5/13/2020	4
T06	Planting	5/13/2020	5/16/2020	4
T07	Mulching	5/17/2020	5/19/2020	3
T08	Edges / Borders	5/20/2020	5/20/2020	1
T09	Trimming / Pruning	5/21/2020	5/21/2020	1
T10	Fertilizing	5/22/2020	5/22/2020	1
T11	Irrigation	5/22/2020	5/24/2020	3
T12	Walkway	5/25/2020	5/25/2020	1
T13	Pond Installation	5/25/2020	5/26/2020	2
T14	Acceptance	5/27/2020	5/27/2020	1

Remarks:
Thirty (30) days warranty starts after acceptance of the project



Free Gantt Chart Template - StudioBinder

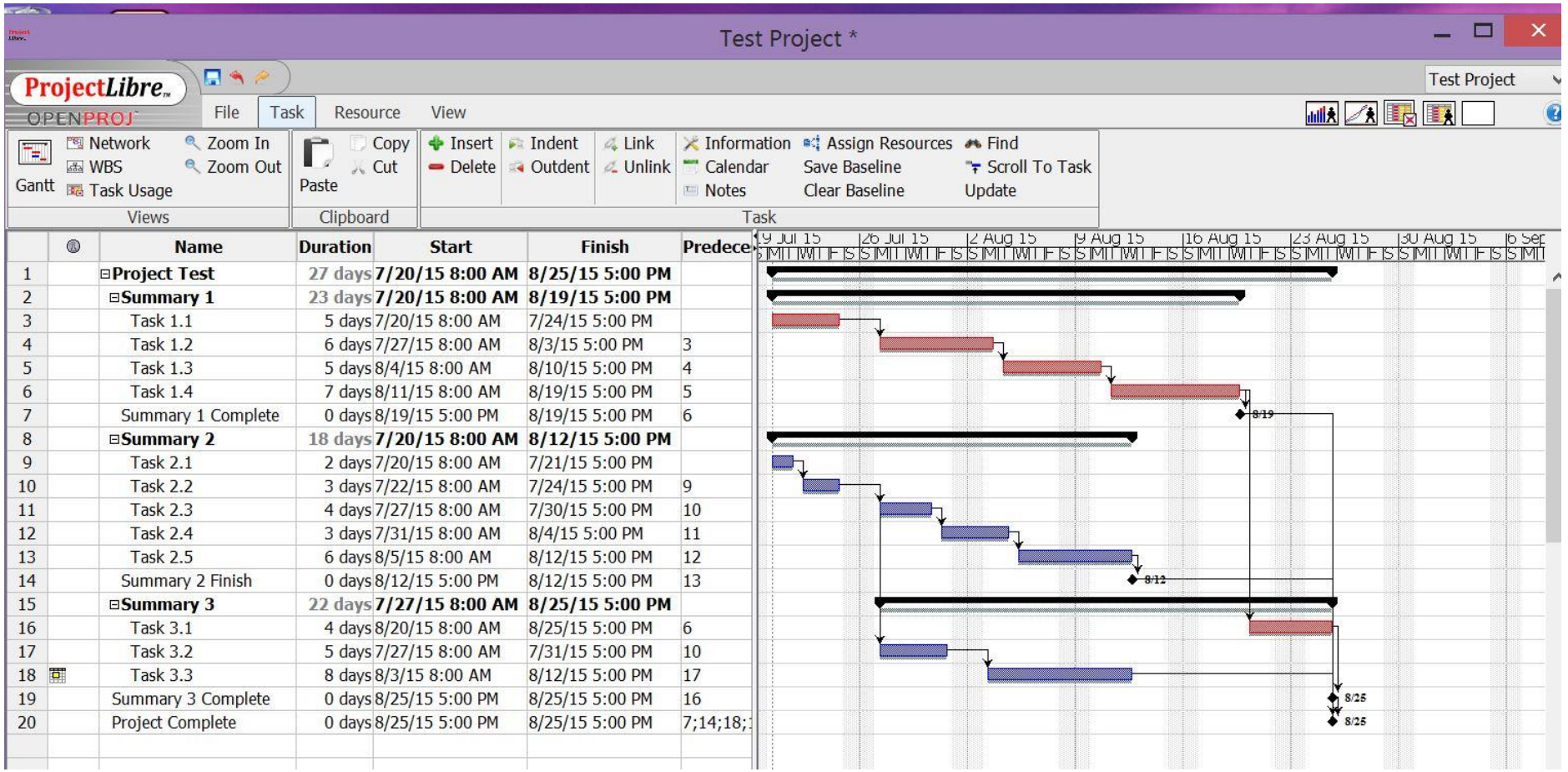
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Shorts
Start Date: 04/01/2017

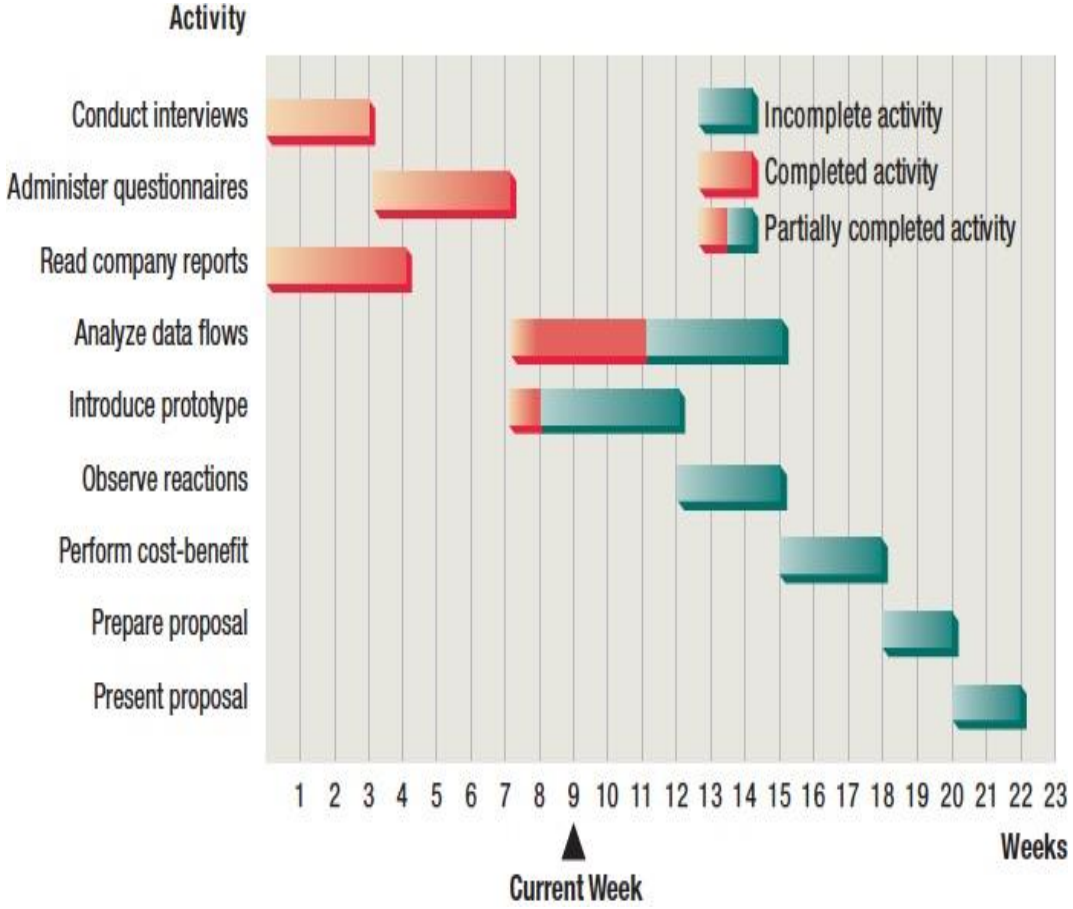
DEVELOPMENT		6 d	
	Finalize Script	L. Kasdan	2 d
	Attachments	L. Wasserman	3 d
	Financing	U. Moneybags	6 d
PRE-PRODUCTION		10 d	
	Casting	M.J. Slater	3 d
	Set Design	B. Lider	3 d
	Locations	G. Places	2 d
	Costumes	H. Boss	3 d
	Hair	V. Sassoon	3 d
	Make-up	M. Factor	3 d
PRODUCTION		7 d	
	Rehearsals	C. DeMille	1 d
	Principal Photography	C. DeMille	5 d
	Second Unit	S. Spielberg	1 d
POST-PRODUCTION		8 d	
	Editing	Q. Tarantino	7 d
	VFX	R. Harryhausen	2 d
	Scoring	J. Williams	2 d
	Sound Editing	B. Osmo	1 d
	Color Grading	M. Gervais	1 d
DISTRIBUTION			
	Distribution Deals	L. Wasserman	
	Marketing Push	L. Wasserman	
	Press Tour	E. Mannix	
	Premieres	E. Mannix	

ANNOTATIONS

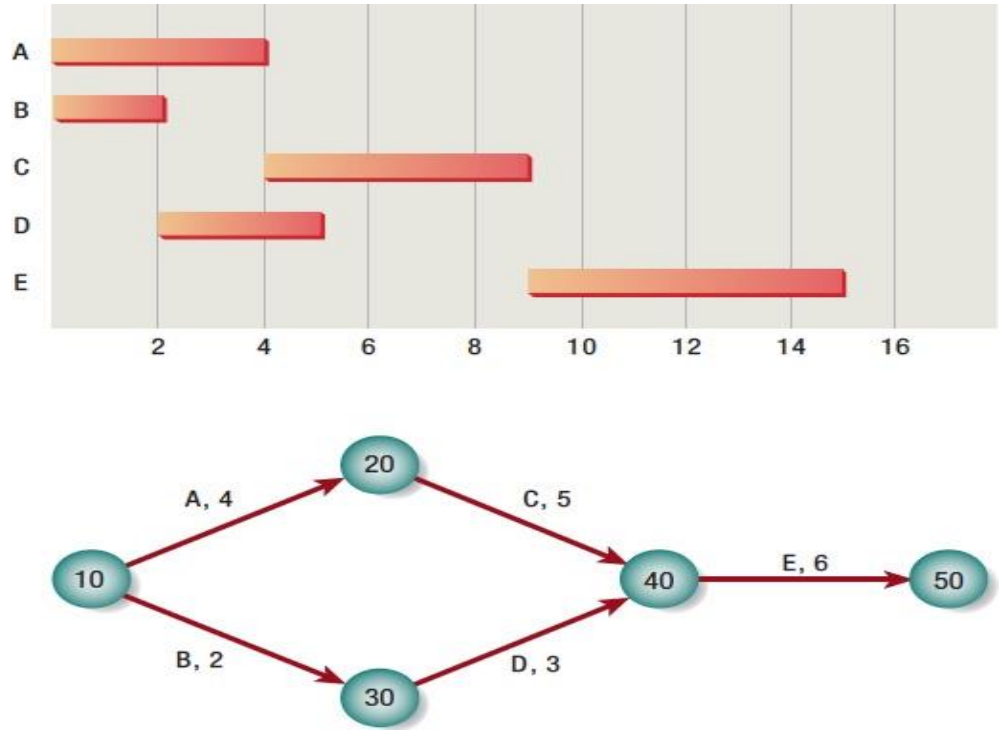
Blocked out event
Blocked out phase
N Dependency
n Dependent Event


Test Project *
ProjectLibre
OPENPROJ
File
Task
Resource
View
Test Project
Gantt
Network
WBS
Task Usage
Zoom In
Zoom Out
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Paste
Copy
Cut
Clipboard
Insert
Delete
Indent
Outdent
Link
Unlink
Information
Calendar
Notes
Assign Resources
Save Baseline
Clear Baseline
Find
Scroll To Task
Update
Task
Name
Duration
Start
Finish
Predece
1 Project Test 27 days 7/20/15 8:00 AM 8/25/15 5:00 PM
2 Summary 1 23 days 7/20/15 8:00 AM 8/19/15 5:00 PM
3 Task 1.1 5 days 7/20/15 8:00 AM 7/24/15 5:00 PM
4 Task 1.2 6 days 7/27/15 8:00 AM 8/3/15 5:00 PM 3
5 Task 1.3 5 days 8/4/15 8:00 AM 8/10/15 5:00 PM 4
6 Task 1.4 7 days 8/11/15 8:00 AM 8/19/15 5:00 PM 5
7 Summary 1 Complete 0 days 8/19/15 5:00 PM 8/19/15 5:00 PM 6
8 Summary 2 18 days 7/20/15 8:00 AM 8/12/15 5:00 PM
9 Task 2.1 2 days 7/20/15 8:00 AM 7/21/15 5:00 PM
10 Task 2.2 3 days 7/22/15 8:00 AM 7/24/15 5:00 PM 9
11 Task 2.3 4 days 7/27/15 8:00 AM 7/30/15 5:00 PM 10
12 Task 2.4 3 days 7/31/15 8:00 AM 8/4/15 5:00 PM 11
13 Task 2.5 6 days 8/5/15 8:00 AM 8/12/15 5:00 PM 12
14 Summary 2 Finish 0 days 8/12/15 5:00 PM 8/12/15 5:00 PM 13
15 Summary 3 22 days 7/27/15 8:00 AM 8/25/15 5:00 PM
16 Task 3.1 4 days 8/20/15 8:00 AM 8/25/15 5:00 PM 6
17 Task 3.2 5 days 7/27/15 8:00 AM 7/31/15 5:00 PM 10
18 Task 3.3 8 days 8/3/15 8:00 AM 8/12/15 5:00 PM 17
19 Summary 3 Complete 0 days 8/25/15 5:00 PM 8/25/15 5:00 PM 16
20 Project Complete 0 days 8/25/15 5:00 PM 8/25/15 5:00 PM 7;14;18;
19 Jul 15
26 Jul 15
2 Aug 15
9 Aug 15
16 Aug 15
23 Aug 15
30 Aug 15
6 Sep
8/19
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8/25

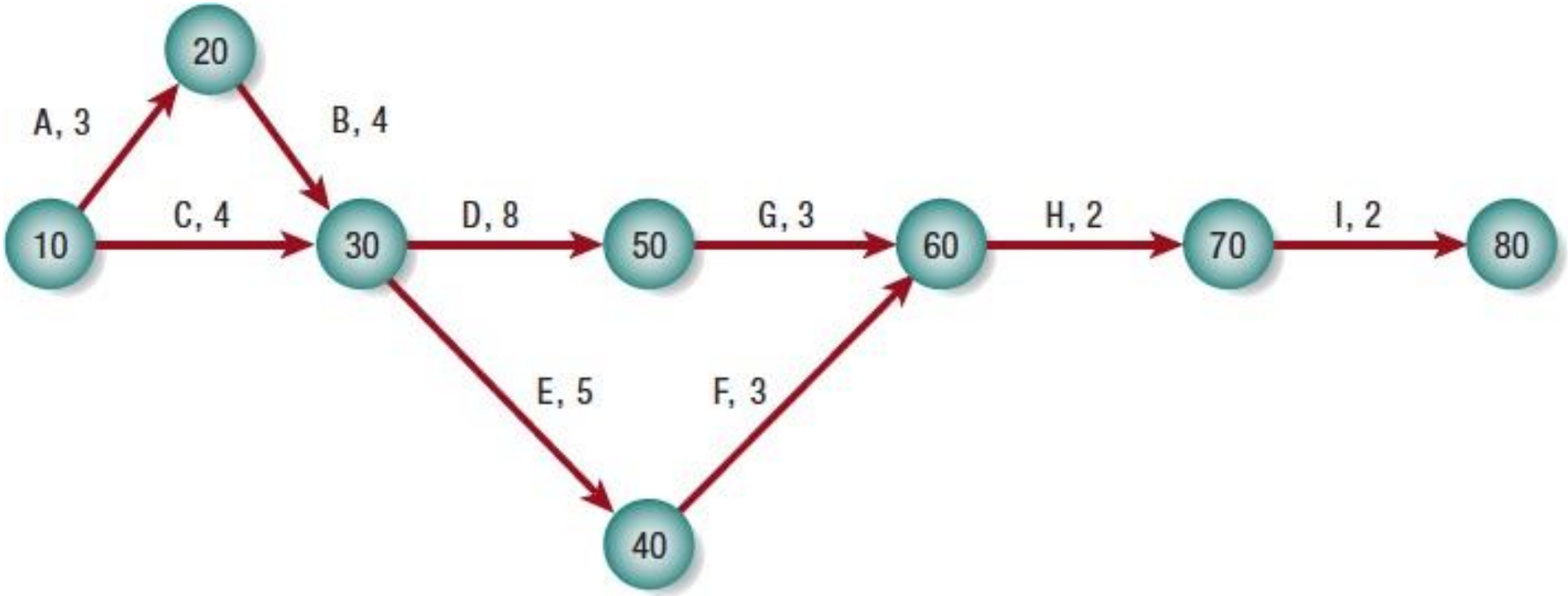
Using a Two-Dimensional Gantt Chart for Planning Activities That Can Be Accomplished in Parallel



A Gantt Chart Compared with a PERT Diagram (Network Diagram) for Scheduling Activities



A Completed PERT Diagram for 、 the Analysis Phase of a Systems Project



PERT Diagram Advantages

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- Easy identification of the order of precedence
- Easy identification of the critical path and thus critical activities
- Easy determination of slack time

3.7 3.7.1 Estimating Costs

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- Using a work breakdown structure
  - Estimate costs for each activity in the work breakdown structure
  - Prepare a budget for the project and have it approved by the organization or client
  - Manage and control the costs throughout the project

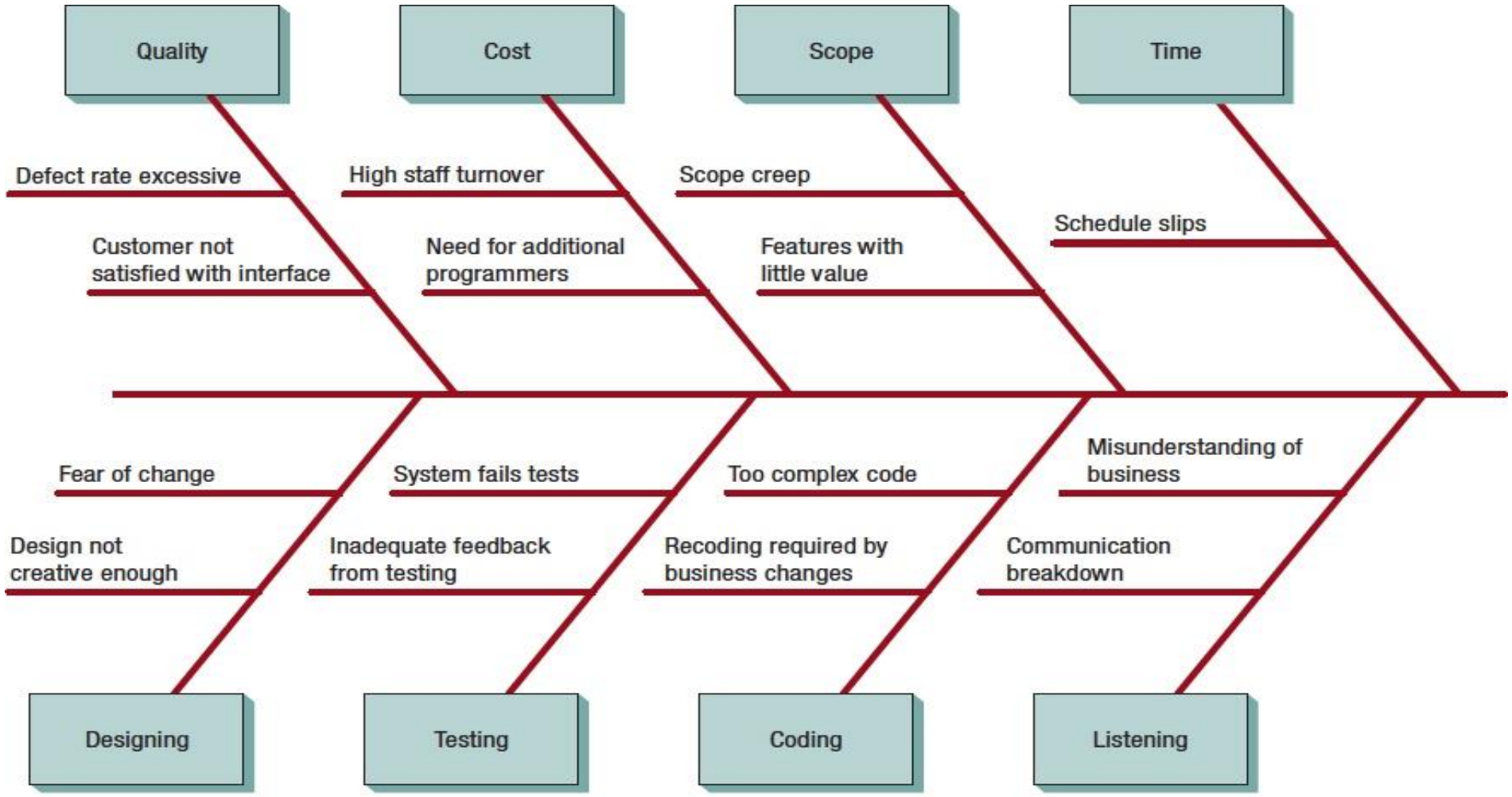
Approaches to Cost Estimation

- Basing estimates on similar projects, also called the top-down approach
- Building bottom-up estimates
- Using parametric modeling

3.7.2 Project Risk

- Project failures may be prevented by:
  - Training
  - Experience
  - Learning why other projects have failed
- Fishbone diagram (or cause-and-effect diagram, or an Ishikawa diagram) systematically lists all of the possible problems that can occur

Fishbone Diagram



3.7.3 Expediting(加速法管理时间)

- **Expediting is speeding up a process**
- **Expediting can help reduce the time it takes to complete an entire project**
- **The expedited activities have to be on the critical path**

Earned Value Management (EVM)

- A technique used to help determine progress (or setbacks) on a project
- Involves:
  - Project cost
  - Project schedule
  - Performance of the project team

Four Key Measures in EVM

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- **Budget at completion (BAC，完工预算成本) is the total budget for the project**
- **Planned value (PV，技术成本) is the value of the work that is to be completed on the project**
- **Actual Cost (AC，实际成本) is the total cost incurred in completing the work**
- **Earned value (EV，挣值) is an estimate of the value of the work performed thus far**

3.8 Managing the Team Project

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- Team management
  - Assembling a team
  - Team communication strategies
  - Project productivity goals
  - Team member motivation

Assembling a Team(组建团队)

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- Shared value of team work
- Good work ethic
- Honesty
- Competency
- Readiness to take on leadership based on expertise
- Motivation
- Enthusiasm for the project
- Trust of teammates

Communication Strategies

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- Teams often have two leaders:
  - Task leader: leads members to accomplish tasks
  - Socioemotional leader: concerned with social relationships
- The systems analyst must manage:
  - Team members
  - Their activities
  - Their time and resources

Project Productivity Goals and Motivation

- Successful projects require that reasonable productivity goals for tangible outputs and process activities be set.
- Goal-setting helps to motivate team members.

Ecommerce Project Management

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- Ecommerce and traditional software project management differences:
  - The data used by ecommerce systems is scattered across the organization
  - Ecommerce systems need a staff with a wide variety of skills
  - Partnerships must be built externally and internally well ahead of implementation
  - Security is of utmost importance

Project Charter（项目章程）

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- **Describes in a written document what the expected results of the systems project are and the time frame for delivery**

Project Charter Clarifies These Questions

- What does the user expect of the project?
- What is the scope of the project?
- What analysis methods will the analyst use to interact with users?
- Who are the key participants?
- What are the project deliverables?
- Who will evaluate the system and how will they evaluate it?
- What is the estimated project timeline?
- Who will train the users?
- Who will maintain the system?

3.9 The Systems Proposal

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- Cover letter
- Title page of project
- Table of contents
- Executive summary
- Outline of systems study with appropriate documentation
- Detailed results of the systems study
- Systems alternatives
- Systems analysts recommendations
- Summary
- Appendices

Using Figures for Effective Communication

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- **Effective use of tables**
- **Effective use of graphs**

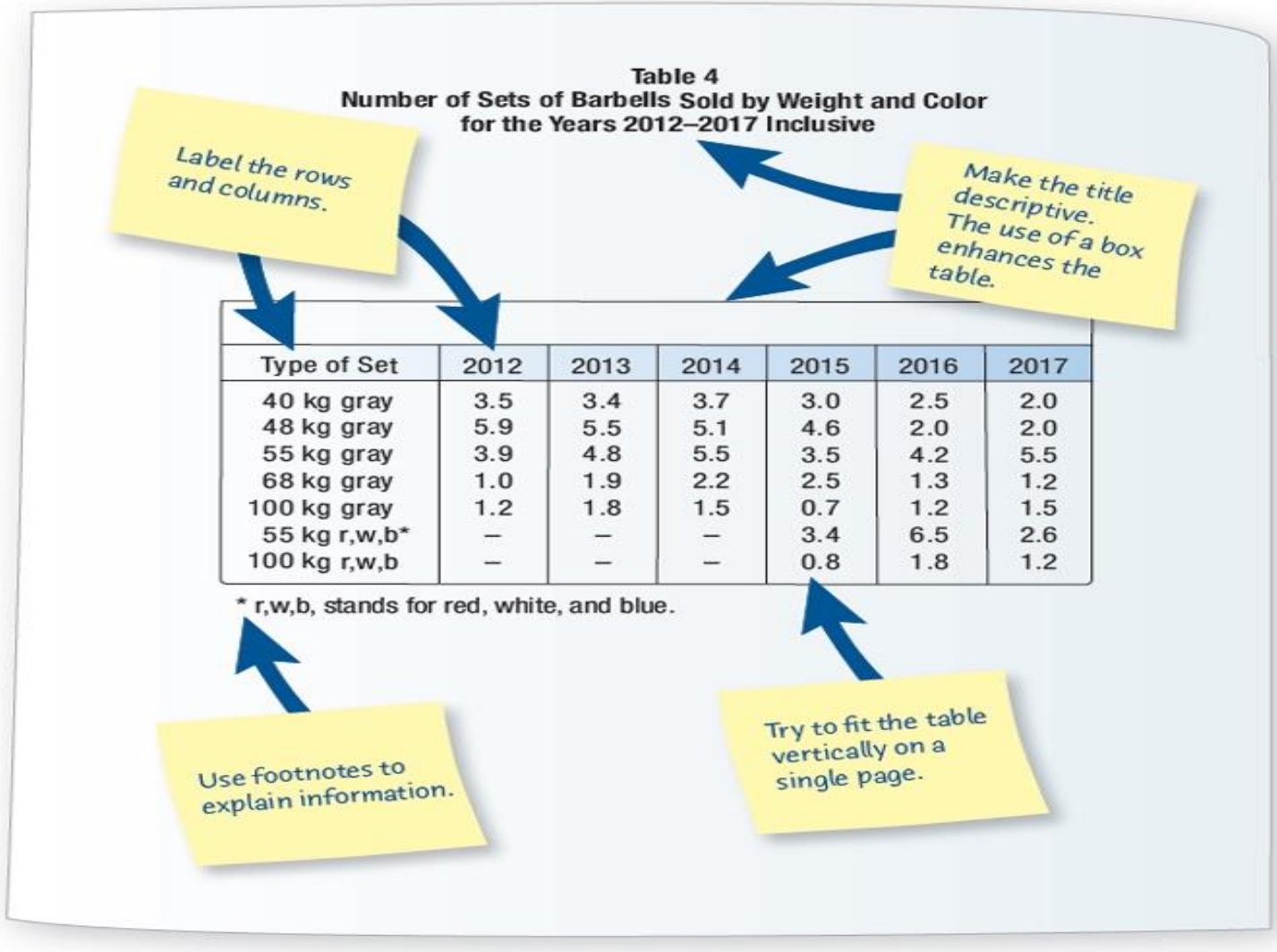
Effective Use of Tables

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- Integrate into the body of the proposal
- Try to fit the entire table vertically on a single page.
- Number and title the table at the top of the page.
- Label each row and column.
- Use a boxed table if room permits.
- Use footnotes if necessary to explain detailed information contained in the table.

Guidelines for Creating Effective Tables

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Effective Use of Graphs

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- Choose a style of graph that communicates your intended meaning well
- Integrate the graph into the body of the proposal
- Give the graph a sequential figure number and a meaningful title
- Label each axis, and any lines, columns, bars, or pieces of the pie on the graph
- Include a key to indicate differently colored lines, shaded bars, or crosshatched areas

Effective Use of Graphs

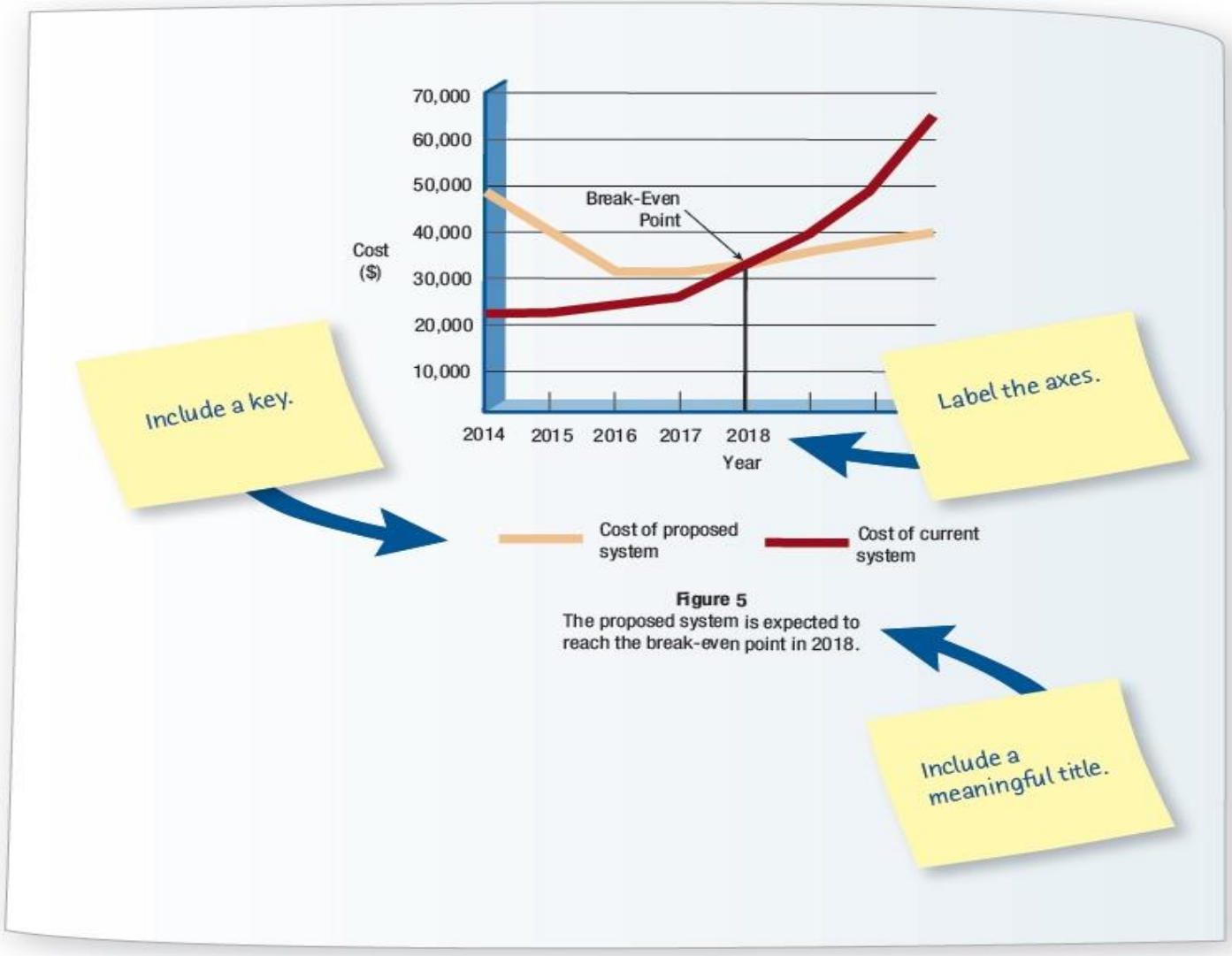


Figure 5
The proposed system is expected to reach the break-even point in 2018.

Summary (1 of 3)

- Project management fundamentals
  - Project initiation
  - Determining project feasibility
  - Activity planning and control
  - Project scheduling
  - Managing systems analysis team members
- Problem definition
  - Issues of the present system
  - The objective for each issue
  - The requirements that must be included in all proposed systems

Summary (2 of 3)

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- Project selection
  - Backed by management
  - Commitment of resources
  - Attains goals
  - Practical
  - Important
- Feasibility
  - Operational
  - Technical
  - Economic

Summary (3 of 3)

- Acquiring hardware and software
- Work breakdown structure
- Project planning
  - Gantt charts
  - PERT
  - Function point analysis
- Team management
- Ecommerce projects
- Preparing a system proposal

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Thank for your Listening.

System Analysis and Design

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